CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 19: Constructions (Circles)

Topic-wise short notes with solved examples

This chapter covers the ruler-and-compass constructions connected with circles: tangents to a circle, and the circumscribed and inscribed circles of a triangle and of a regular hexagon.

1 Rules for every construction

- Use only a ruler and a pair of compasses, with a sharp pencil.
- Draw a rough sketch first and mark the given measurements on it.
- Leave all construction arcs and lines visible; they earn marks.
- Finish by measuring and writing down whatever the question asks for.
- The two basic constructions used again and again are the **perpendicular bisector** of a line segment and the **bisector of an angle**.

2 Tangent at a point on the circle

- **Given**: a circle with centre O and a point P on it.
- Step 1: join OP.
- Step 2: at P, construct a line APB perpendicular to OP.
- APB is the required tangent.
- **Why it works**: the tangent at any point is perpendicular to the radius through that point.

3 Tangents from an external point

- **Given**: a circle with centre O and a point P outside it.
- Step 1: join OP and draw its perpendicular bisector to find the mid-point M of OP.
- Step 2: with centre M and radius MO, draw a circle. Let it cut the given circle at A and B.
- Step 3: join PA and PB. These are the two tangents.
- **Why it works**: OP is a diameter of the second circle, so $\angle OAP = 90^\circ$ (angle in a semicircle). Thus PA is perpendicular to the radius OA.

Example 3.1 A circle has radius 3 cm and a point P is 5 cm from its centre. Write the steps to construct the tangents from P, and calculate their length.

SOLUTION

Step 1: draw a circle with centre O and radius 3 cm. Mark P with $OP = 5$ cm.

Step 2: bisect OP at M. With centre M and radius $MO = 2.5$ cm draw a circle cutting the first circle at A and B.

Step 3: join PA and PB.

Length: $PA = \sqrt{OP^2 - OA^2} = \sqrt{25 - 9}$

$PA = PB = 4$ **cm (measure to confirm).**

REMEMBER

Always check a tangent construction by calculation: the measured length should match $\sqrt{d^2 - r^2}$.

4 Circumscribed circle of a triangle

- The **circumcircle** passes through all three vertices. Its centre is the **circumcentre** and its radius the **circumradius**.
- Step 1: construct the triangle.
- Step 2: draw the perpendicular bisectors of any two sides. Let them meet at O.
- Step 3: with centre O and radius OA, draw a circle. It passes through A, B and C.
- **Why it works**: every point on the perpendicular bisector of a side is equidistant from its two end points, so $OA = OB = OC$.

Triangle	Position of the circumcentre
Acute-angled	Inside the triangle
Right-angled	The mid-point of the hypotenuse
Obtuse-angled	Outside the triangle

Example 4.1 A right-angled triangle has sides 6 cm, 8 cm and 10 cm. Where is its circumcentre, and what is its circumradius?

SOLUTION

The circumcentre of a right-angled triangle is the mid-point of the hypotenuse.

It is the mid-point of the 10 cm side, and the circumradius is 5 cm.

5 Inscribed circle of a triangle

- The **incircle** touches all three sides. Its centre is the **incentre** and its radius the **inradius**.
- Step 1: construct the triangle.
- Step 2: draw the bisectors of any two angles. Let them meet at I.
- Step 3: from I draw a perpendicular ID to any one side.

- Step 4: with centre I and radius ID, draw a circle. It touches all three sides.
- **Why it works**: every point on the bisector of an angle is equidistant from the arms of the angle, so I is equidistant from the three sides.
- The incentre always lies inside the triangle.

Example 5.1 An equilateral triangle has side 6 cm. Calculate its inradius and its circumradius.

SOLUTION

Height $= \frac{\sqrt{3}}{2} \times 6 = 3\sqrt{3}$ cm. In an equilateral triangle the incentre and circumcentre coincide at the centroid, which divides the height in the ratio $2 : 1$.

Circumradius $= \frac{2}{3} \times 3\sqrt{3} = 2\sqrt{3}$ cm and inradius $= \frac{1}{3} \times 3\sqrt{3} = \sqrt{3}$ cm

Inradius ≈ 1.73 cm; circumradius ≈ 3.46 cm

6 Circles and a regular hexagon

- A regular hexagon has six equal sides and each interior angle is $120°$.
- **To construct a regular hexagon of side** a: draw a circle of radius a, and with the same radius step off six arcs round the circle. Join the points in order.
- **Circumscribed circle** of a regular hexagon: draw the perpendicular bisectors of any two sides; they meet at the centre O. The radius is OA, and it **equals the side** of the hexagon.
- **Inscribed circle** of a regular hexagon: use the same centre O. Draw the perpendicular from O to any side; its length is the radius, equal to $\frac{\sqrt{3}}{2}a$.

Example 6.1 A regular hexagon has side 4 cm. Find the radius of its circumscribed circle and of its inscribed circle.

SOLUTION

Circumradius $=$ side $= 4$ cm

Inradius $= \frac{\sqrt{3}}{2} \times 4 = 2\sqrt{3}$ cm

4 cm and about 3.46 cm

7 Common mistakes

- Using perpendicular bisectors for the incircle, or angle bisectors for the circumcircle.
- Taking the radius of the incircle as the distance from I to a vertex. It is the **perpendicular** distance to a side.
- Drawing a tangent by eye instead of constructing it.
- Rubbing out the arcs.